HOMEWORK 14

Due date: Monday of Week 15,
Exercises: 2.1, 2.2, 2.3, 2.7, 2.13, 2.14, 2.17, 2.18, 3.2, 3.3, 5.5, 5.7, 5.11, 5.12, 6.1, 6.2, 6.4, 6.5 pages 221-223.

Problem 1. *Let G be a finite group with $|G| = n$. Let $\iota : G \to S_n$ be the embedding in Cayley's theorem (namely, $\iota : G \to \mathrm{Perm}(G)$ is determined by the left multiplication action $G \times G \to G$, $(g, x) \mapsto gx$).*

(1) *Let $h \in G$ be an element of order d (so that $d|n$). Show that $\iota(h)$ is a product of n/d disjoint cycles of length d.*
(2) *Suppose $n = 4m + 2$. Show that $\iota(h) \in A_n$ if and only if the order of h is odd.*

Hint: Let $H = \langle h \rangle \subset G$ and consider the right coset decomposition $G = \coprod Hg_i$. For (2), see Ex. 10.1, page 74. This problem was taken from a Quora answer here.

Proof. We consider part (2). If h has odd order, then $\iota(h)$ has odd order and thus $\iota(h) \in A_n$ by Example 2.10.3. Here are some details. Consider the map sign : $S_n \to \{\pm 1\}$ and its restriction $\phi : \langle \iota(h) \rangle \to \{\pm 1\}$. Since $|\mathrm{Im}(\phi)||2$ and $|\mathrm{Im}(\phi)|$ divides $|\langle \iota(h) \rangle|$, we get $\mathrm{Im}(\phi) = 1$. Thus $h \in \ker(\phi) = A_n$. This direction is true without any other condition. Conversely, suppose that $n = 4m + 2$ and $\iota(h) \in A_n$, we need to show that the *order*$(h) = d$ is odd. We know $\iota(h)$ is a product of n/d disjoint cycles and thus $\iota(h) \in A_n$ if and only if $n - n/d$ is even, (see the next paragraph for an explanation of this and this is exactly Ex 10.1, page 74). Thus $\iota(h) \in A_n$ implies that n/d is even and thus $d|(2m + 1)$, which is odd. $\square$

Problem 2. *Let C be a conjugacy class of S_n. Decompose $C \cap A_n$ as conjugacy classes of A_n.*

This is roughly Ex.5.11. For future references, here is a more precise statement. Suppose that $\sigma \in S_n$ has cycle lengths $k_1, \ldots, k_m$ with $k_1 + \cdots + k_m = n$. This means that σ can be written as a product of m disjoint cycles of lengths $k_1, \ldots, k_m$. For example, for $\sigma = (123)(45)(6) \in S_6$, we have $m = 3$, $k_1 = 3, k_2 = 2, k_3 = 1$. Using this notation, we have $\mathrm{sign}(\sigma) = (-1)^{k_1 - 1 + k_2 - 1 + \cdots + k_m - 1} = (-1)^{n-m}$. Thus σ is an even permutation (namely, $\sigma \in A_n$) if and only if $n - m$ is even. For example, $(123)(45)(6) \in S_6$ has signature -1. This is Ex 10.1, page 74. Now suppose that $n - m$ is even and so that $\sigma \in A_n$. We consider the S_n conjugacy class $C(\sigma) = \{g\sigma g^{-1} : g \in S_n\}$.

Problem 3. (1) *If all k_i are odd and distinct, then $C(\sigma)$ is the union of two A_n conjugacy classes and these two conjugacy classes have the same order.*
(2) *Otherwise (which means, either one of k_i is even, or there are at least two k_i are the same), then $C(\sigma)$ is still a single A_n-conjugacy class.*

For example, in S_4, if $\sigma = (12)(34)$, then $C(\sigma)$ is a single A_4 conjugacy class; if $\sigma = (123)(4)$, then $C(\sigma)$ is the union of two different A_4 conjugacy classes. Actually, one can see that (123) and (132) are not conjugate in A_4. Write

$$\sigma = (i_{11} \ldots i_{1k_1}) \ldots (i_{m1} \ldots i_{mk_m})$$

be the disjoint cycle decomposition of σ. Denote

$$C_1(\sigma) = \left\{g\sigma g^{-1} : g \in A_n\right\}, \quad C_2(\sigma) = \left\{g\sigma g^{-1} : g \in S_n - A_n\right\}.$$

Then $C(\sigma) = C_1(\sigma) \cup C_2(\sigma)$ and both $C_i(\sigma), i = 1, 2$ are A_n-conjugacy classes. Thus $C(\sigma)$ is at most two A_n-conjugacy classes. Moreover, $C(\sigma)$ is still a single conjugacy A_n class iff $C_1(\sigma) \cap C_2(\sigma) \neq \emptyset$ iff $\sigma \in C_2(\sigma)$ iff $\sigma = \tau\sigma\tau^{-1}$ for some odd permutation τ. Suppose one of k_j is even in the decomposition of σ. Let $\tau = (i_{j1} \ldots i_{jk_j})$, then τ is odd and $\tau\sigma\tau^{-1} = \sigma$ and thus $C_1(\sigma) = C_2(\sigma)$. Suppose that all k_i are odd and two of them are equal. Without loss of generality, we assume that $k_1 = k_2$. Consider

$\tau = (i_{11}i_{21})(i_{12}i_{22})\dots(i_{1k_1}i_{2k_2})$, which is a product of $k_1 = k_2$ transpositions and thus is odd. We have $\tau\sigma = \sigma\tau$. On the first k_1 indices, this can be seen from the following diagram

$$\begin{tikzcd}
i_{11} \arrow[r, "\sigma"] \arrow[d, "\tau"] & i_{12} \arrow[r, "\sigma"] \arrow[d, "\tau"] & \cdots \arrow[r, "\sigma"] \arrow[d] & i_{1k_1} \arrow[r, "\sigma"] \arrow[d, "\tau"] & i_{11} \arrow[d, "\tau"] \\
i_{21} \arrow[r, "\sigma"] & i_{22} \arrow[r, "\sigma"] & \cdots \arrow[r, "\sigma"] & i_{2k_2} \arrow[r, "\sigma"] & i_{21}
\end{tikzcd}$$

This proves the second claim. Now we assume that all k_i are odd and distinct and we need to show that there is no odd permutation τ such that $\tau\sigma = \sigma\tau$. Suppose that $\tau \in S_n$, then $\tau\sigma\tau^{-1}$ is a permutation with the same cycle lengths as σ. Since each cycle length are distinct, the condition $\tau\sigma\tau^{-1}$ implies that τ preserves each cycles in σ, namely,

$$\tau(i_{j1}i_{j2}\dots i_{jk_j})\tau^{-1} = (i_{j1}i_{j2}\dots i_{jk_j})$$

for each j. Thus we have a commutative diagram

$$\begin{tikzcd}
i_{j1} \arrow[r, "\sigma"] \arrow[d, "\tau"] & i_{j2} \arrow[r, "\sigma"] \arrow[d, "\tau"] & \cdots \arrow[r, "\sigma"] \arrow[d] & i_{jk_j} \arrow[r, "\sigma"] \arrow[d, "\tau"] & i_{j1} \arrow[d, "\tau"] \\
\tau(i_{j1}) \arrow[r, "\sigma"] & \tau(i_{j2}) \arrow[r, "\sigma"] & \cdots \arrow[r, "\sigma"] & \tau(i_{jk_j}) \arrow[r, "\sigma"] & \tau(i_{j1})
\end{tikzcd}$$

for each j. This means that $\tau : \{i_{j1}, \dots, i_{jk_j}\} \to \{i_{j1}, \dots, i_{jk_j}\}$ is a cycle of length k_j determined by $\tau(i_{j1})$. From this we see that $\tau = (i_{j1}\dots i_{jk_j})^{a_j}$ for some $a_j \in \mathbb{Z}$ for the j-th part. Thus

$$\tau = (i_{11}\dots i_{1k_1})^{a_1}\dots(i_{m1}\dots i_{km})^{a_m}.$$

This τ is even because k_j is odd for each j. Thus there is no odd permutation τ such that $\tau\sigma = \sigma\tau$. This proves (1).

Problem 4. *Given an element $p \in S_n$ with m_1 1-cycles, ..., m_n n-cycles. So $\sum_{i=1}^n im_i = n$. For example, for the cycle $p = (123)(45)(67)(89)$ of S_{10}, we have $m_1 = 1, m_2 = 3, m_3 = 1$ and $m_i = 0$ for $i \geq 4$. Determine how many elements are in the conjugacy class determined by p.*

Answer:

$$|C| = \frac{n!}{1^{m_1}m_1!2^{m_2}m_2!\dots n^{m_n}m_n!} = \frac{n!}{\prod_{i=1}^n i^{m_i}m_i!}.$$

For example, in S_5, the conjugacy class contains $(12)(345)$ has $\frac{5!}{2^1\cdot 1!\cdot 3^1\cdot 1!} = 20$ elements, and the conjugacy class contains (12345) has $\frac{5!}{5^1} = 4! = 24$ elements.

Problem 5. *Let G be a finite group, H be a subgroup of G. Let $C \subset G$ be a conjugacy class and suppose*

$$H \cap C = \coprod_{i=1}^r D_i,$$

where each D_i is a conjugacy class of H. Consider the set

$$X_i = \{(c, g) \in C \times G : g^{-1}cg \in D_i\}.$$

Express $|X_i|$ in terms of $|G|, |H|, |D_i|$.

Hint: Consider the group action $G \times X_i \to X_i$ defined by $x.(c, g) = (xcx^{-1}, xg)$. For $d \in D_i$, we consider $(d, 1) \in X_i$ and its G-orbit $O((d, 1))$. If $(c, g) \in X_i$, then $(c, g) = g.(g^{-1}cg, 1)$ and $g^{-1}cg \in D_i$. Thus $X_i = \cup_{d\in D_i} O((d, 1))$. Moreover, if $O((d, 1)) = O((d', 1))$, then $d = d'$. Thus $X_i = \coprod_{d\in D_i} O((d, 1))$. Consider the G action on $O((d, 1))$ we have $Stab((d, 1)) = 1$. Thus $|O((d, 1))| = |G|$ and $|X_i| = |G||D_i|$.

Problem 6. *Let $G = D_4 = \{1, x, x^2, x^3, y, xy, x^2y, x^3y\}$ with $x^4 = 1 = y^2, yxy^{-1} = x^3$ and $H = \{1, x^2, y, x^2y\} \subset G$. Find all conjugacy classes C of G, and for each conjugacy class C of G, decompose $C \cap H$ into conjugacy classes of H.*

Problem 7. *Let $G = \mathrm{GL}_2(\mathbb{F}_p), H = \mathrm{SL}_2(\mathbb{F}_p) = \{g \in G : \det(g) = 1\}$. Let $C \subset G$ be the conjugacy class of the element $u = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Namely,*

$$C = \left\{gug^{-1} : g \in G\right\}.$$

Try to decompose $C \cap H$ into conjugacy classes of H.

We have the decomposition

$$\mathrm{GL}_2(\mathbb{F}_p) = \coprod_{a \in \mathbb{F}_p^\times} \mathrm{SL}_2(\mathbb{F}_p) d_a, \quad d_a = \begin{bmatrix} a & \\ & 1 \end{bmatrix}.$$

For each $a \in \mathbb{F}_p^\times$, let

$$C_a = \left\{gug^{-1}, g \in \mathrm{SL}_2(\mathbb{F}_p) d_a\right\}.$$

Then each C_a is an $\mathrm{SL}_2(\mathbb{F}_p)$ conjugacy class and $C = \cup_{a \in \mathbb{F}_p^\times} C_a$. We need to determine when $C_a = C_b$. Claim: $C_a = C_b$ iff $ab^{-1} = x^2$ for some $x \in \mathbb{F}_p^\times$. If $a = bx^2$, we have

$$\begin{bmatrix} 1 & a \\ & 1 \end{bmatrix} = \begin{bmatrix} x & \\ & x^{-1} \end{bmatrix} \begin{bmatrix} 1 & b \\ & 1 \end{bmatrix} \begin{bmatrix} x^{-1} & \\ & x \end{bmatrix}.$$

Thus $d_a u d_a^{-1} \in C_b$ and thus $C_a = C_b$. Conversely, suppose that $C_a = C_b$, there exists $g = \begin{bmatrix} x & y \\ z & w \end{bmatrix} \in \mathrm{SL}_2(\mathbb{F}_p)$ such that

$$d_a u d_a^{-1} = g d_b u d_b^{-1} g^{-1},$$

or

$$\begin{bmatrix} 1 & a \\ & 1 \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & y \\ z & w \end{bmatrix} \begin{bmatrix} 1 & a \\ & 1 \end{bmatrix}.$$

We can solve that $z = 0$ and $w = ax/b$. Thus $\det(g) = 1$ implies that $ab^{-1}x^2 = 1$ or $ab^{-1} = x^{-2}$. Consider the map

$$\phi : \mathbb{F}_p^\times \to \mathbb{F}_p^\times, \phi(x) = x^2.$$

Assume p is odd. Then $\ker(\phi) = \{\pm 1\}$ and thus $|\mathrm{Im}(\phi)| = (p-1)/2$. Take $\kappa \in \mathbb{F}_p^\times - \mathrm{Im}(\phi)$. Thus $\mathbb{F}_p^\times = \mathrm{Im}(\phi) \coprod \kappa \mathrm{Im}(\phi)$. Thus

$$C = C_1 \coprod C_\kappa.$$

Problem 8. *Let p be a prime number. Show that the cyclic group C_{p^n} for $n \geq 2$ is not a semi-direct product of two proper subgroups.*

If $G = C_{p^n}$ is semidirect product of H and K it must be direct product of H and K because G is abelian. But there is no element in $H \times K$ of order p^n.

Proposition 7.3.3, page 198, says that every group of order p^2 is abelian. In the following, we give some examples of non-abelian group of order p^3. We assume that $p > 2$. If $p = 2$, we have seen that the quaternion group is an order 2 non-abelian group.

The first one is called Heisenberg group of the field $\mathbb{F}_p$, and we temporarily denote it by $H(\mathbb{F}_p^2)$. (This looks like a wired notation, but it has generalizations). It is defined by

$$H(\mathbb{F}_p^2) = \left\{ \begin{bmatrix} 1 & x & z \\ & 1 & y \\ & & 1 \end{bmatrix} \in \mathrm{GL}_3(\mathbb{F}_p), x, y, z \in \mathbb{F}_p \right\}.$$

Its group structure is defined by matrix multiplication. The other group is temporarily denoted by G_p and it is defined by

$$G_p = \left\{ \begin{bmatrix} x & y \\ & 1 \end{bmatrix} \in \mathrm{GL}_2(\mathbb{Z}/p^2\mathbb{Z}) : x \equiv 1 \bmod p, y \in \mathbb{Z}/p^2\mathbb{Z} \right\}.$$

Problem 9. *Show that $H(\mathbb{F}_p^2)$ and G_p are non-abelian group of order p^3. Moreover, show that they are indeed semidirect products of their own subgroups.*

It can be shown that these are the only two non-abelian groups of order p^3 up to isomorphism.

Problem 10. *Determine the class equations of* $D_n, n \geq 3$, $\mathrm{GL}_2(\mathbb{F}_p), \mathrm{SL}_2(\mathbb{F}_p)$ *and* $\mathrm{PSL}_2(\mathbb{F}_p)$, *where* $\mathrm{PSL}_2(\mathbb{F}_p) = \mathrm{SL}_2(\mathbb{F}_p)/\{\pm I_2\}$.

If you find this hard, at least try some examples for small p.
If $n = 2k+1$ is odd, then the conjugacy classes in D_n are

$$\{1\}, \{x, x^{-1}\}, \ldots, \{x^k, x^{-k}\}, \{y, x^2y, x^{2m}y, m \in \mathbb{Z}\} = \{y, xy, \ldots, x^{2k}y\}.$$

So the class equation is $n = 1 + 2 + \cdots + 2 + (2k+1)$. If $n = 2k$ is even, then the conjugacy classes are

$$\Big\{\{1\}, \{x^k\}, \{x, x^{-1}\}, \ldots, \Big\{x^{k-1}, x^{-(k-1)}\Big\}, \{y, x^2y, \ldots, x^{2k-2}y\}, \{xy, x^3y, \ldots, x^{2k-1}y\}\Big\}.$$

So the class equation is

$$n = 1 + 1 + 2 + \cdots + 2 + k + k$$

with $(k-1)$s 2.

For $\mathrm{GL}_2(\mathbb{F}_p)$. The conjugacy classes are given in HW 3. For $g = \begin{bmatrix} a & \\ & a \end{bmatrix}$, we have $|C(g)| = 1$ and there are $p-1$ such conjugacy classes. If $g = \begin{bmatrix} a & 1 \\ & a \end{bmatrix}$, then its centralizer is $Z(g) = \left\{\begin{bmatrix} x & y \\ & x \end{bmatrix}\right\}$. Thus $|Z(g)| = p(p-1)$ and $|C(g)| = |G|/|Z(g)| = p^2 - 1$. There are $p-1$ such classes. If $g = \begin{bmatrix} a & \\ & b \end{bmatrix} = t(a,b)$ with $a \neq b$, then $Z(g)$ is the diagonal subgroup and thus $|Z(g)| = (p-1)^2$ and $|C(g)| = p(p+1)$. There are $(p-1)(p-2)/2$ such classes because $t(b,a) \in C(g)$. Finally, fix $\kappa \in \mathbb{F}_p^\times - \mathbb{F}_p^{\times,2}$ and consider $g = \begin{bmatrix} a & b\kappa \\ b & a \end{bmatrix} = e(a,b)$ for $a \in \mathbb{F}_p, b \in \mathbb{F}_p^\times$, then

$$Z(g) = \left\{\begin{bmatrix} x & \kappa z \\ z & x \end{bmatrix}, x^2 - \kappa z^2 \neq 0.\right\}$$

Then $|Z(g)| = (p^2-1)$ and thus $|C(g)| = p^2 - p$. There are $p(p-1)/2$ such classes because $e(a,-b) \in C(g)$. Thus the class equation for $\mathrm{SL}_2(\mathbb{F}_p)$ is

$$(p^2-1)(p^2-p) = (p-1) + (p-1)(p^2-1) + p(p+1)(p-1)(p-2)/2 + (p^2-p)p(p-1)/2.$$

Problem 11. *Let G be a finite group and let p be the smallest prime that divides $|G|$.*

(1) *If H is a normal subgroup of G such that $|H| = p$, show that $H < Z(G)$.*
(2) *If $H < G$ is a subgroup such that $[G : H] = p$, then H is a normal subgroup of G.*

Part (1) is Ex 6.4 page 223. and part (2) generalizes Ex.8.10, page 73. For (1), Consider the conjugation action of G on H, which gives a homomorphism $\phi : G \to \mathrm{Aut}(H) = (\mathbb{Z}/p\mathbb{Z})^\times$. Thus $\mathrm{Im}(\phi) = 1$. For (2), Consider the action of H on $X = G/H$. Suppose that $X = \cup_{k=1}^m O_{x_k}$ be the disjoint union of orbits. We have

$$p = |G/H| = \sum |O_{x_k}|.$$

Note that $|O_{x_k}| | H$, which also divides $|G|$. Since p is the smallest prime which divides G, we get either $|O_{x_k}| = 1$ or $|O_{x_k}| \geq p$. But there is at least one x_k such that $|O_{x_k}| = 1$ (the orbit $[H]$), we get $|O_{x_k}| = 1$ for each k. This means that for any coset gH, we have $hgH = gH$ for any $h \in H$. Thus $ghg^{-1} \in H$ for every $g \in G, h \in H$. This shows that H is normal in G. Here is a second way to prove (2). Consider the action of G on $X := G/H$ by left multiplication. We get a group homomorphism $\phi : G \to Perm(X) = S_p$. We first claim that $K := \mathrm{Ker}(\phi) < H$. In fact, if $k \in K$, then $\phi(k) = \mathrm{id}_X$, which means that $kgH = gH$ for any $g \in G$. In particular, we get $kH = H$ and thus $k \in H$. Secondly, we have $G/K \cong \mathrm{Im}(\phi) < S_p$. Thus $[G : K] | p!$ and $[G : K] || G|$. We get $[G : K] | gcd(p!, |G|)$. Since p is the smallest prime that divides $|G|$, we get $[G : K] | p$. Thus $[G : K] = 1$ or p. If $[G : K] = 1$, then $G = K$ and thus $G = K < H$, which is impossible. Thus $[G : K] = p = [G : H]$. The condition $K < H$ implies that $H = K$, which is normal.

The following is a solution of Exercise 6.2, page 223.

Proof. We only show $B = N(B)$ and the proof of the other equation is the same. The inclusion $B \subset N(B)$ is obvious. Suppose $g \in N(B)$, we need to show $g \in B$. Using the Bruhat decomposition, we write $g = bP_\sigma b'$. The condition implies that $gBg^{-1} = B$, which is equivalent to $P_\sigma B P_\sigma^{-1} = B$. We need to show that $\sigma = 1$. Suppose $\sigma \neq 1$, and thus σ contains a nontrivial cycle, say $(i_1 i_2 \dots i_k)$ with $i_1 < i_2 < \cdots < i_k$. To easy the notation, we set $l = i_{k-1}, m = i_k, j = i_1$, then we have $\sigma(l) = m > l$ and $\sigma(m) = j < m$. We consider the matrix $b \in B_n$ such that

$$b_m = \epsilon_l + \epsilon_m; b_k = \epsilon_k, k \neq m.$$

Here b_m denotes the m-th column of b. In other words, b is the matrix of the form

$$\begin{bmatrix} 1 & & & & \\ & 1 & & 1 & \\ & & 1 & & \\ & & & 1 & \\ & & & & 1 \end{bmatrix}$$

where the 1 on the upper right is at the l-th row and m-th column. This b is indeed upper triangular because $l < m$. The assumption says that $P_\sigma b P_\sigma^{-1} = b' \in B_n$. Thus

$$P_\sigma b = b' P_\sigma.$$

In particular, we have $P_\sigma b \epsilon_m = b' P_\sigma \epsilon_m$. We have

$$P_\sigma b \epsilon_m = P_\sigma b_m = P_\sigma(\epsilon_l + \epsilon_m) = \epsilon_{\sigma(l)} + \epsilon_{\sigma(m)} = \epsilon_m + \epsilon_j.$$

On the other hand, we have

$$b' P_\sigma \epsilon_m = b' \epsilon_{\sigma(m)} = b'_j.$$

Thus we get

$$b'_j = \epsilon_m + \epsilon_j$$

which contradicts to that b' is upper triangular because $m > j$. $\square$